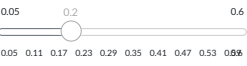


Undesirable response rate (p_0)



Acceptable response rate (p_1)



Type I error (α)



Power ($1 - \beta$)



Design Criterion

☒ Optimal ☐ MiniMax

Beta Prior on Response Rate

$Beta(\alpha, \beta)$ prior on p . Default $Beta(1, 1)$ is uniform.

α (prior successes)

1

β (prior failures)

1

Simon Design

Posterior Distribution

Total observations (n)



Observed successes (a)



$P(p \leq p_0 \mid \text{data})$

0.029

$P(p \geq p_1 \mid \text{data})$

0.186

Posterior: Beta(13.0, 26.0) — 12 successes in 37 observations (observed rate = 32.4%)

